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[23CSE107]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY-2024
INTRODUCTION TO COMPUTER ARCHITECTURE

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define multiplexer and demultiplexer?	L-1	CO1	**02**
	b.	Convert the given number $(1011)_2$ to Decimal	L-5	CO6	**02**
	c.	Write any one data manipulation instruction	L-2	CO3	**02**
	d.	Why Interrupt is important	L-3	CO4	**02**
	e.	How Hit Ratio calculated in cache memory?	L-4	CO5	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Describe the basic operational concepts of a computer system, including memory, input/output, and processing	L-2	CO1	**05**
		OR			
	b.	Explain the concept of a bus in a computer system and describe its basic operation	L-2	CO1	**05**
3.	**a.**	Illustrate the arithmetic Micro Operations and Logic Micro Operations in detail.	L-5	CO2	**05**
		OR			
	b.	Explain in detail about different types of Error Detection Codes	L-2	CO2	**05**
4.	**a.**	Difference between RISC and CISC? Explain in detail.	L-4	CO3	**05**
		OR			
	b.	Explain different types of Instruction format in detail.	L-3	CO3	**05**
5.	**a.**	Differentiate serial and parallel data transfer.	L-4	CO4	**05**
		OR			
	b.	Explain different types of Pipelining.	L-2	CO4	**05**
6.	**a.**	Define Cache Memory. Explain Cache Memory with suitable examples.	L-2	CO5	**05**
		OR			
	b.	Explain in detail about Mapping Functions.	L-3	CO5	**05**
7.	**a.**	Illustrate the different types of binary operations with example.	L-5	CO6	**05**

		OR			
	b.	Compute the following i. $(0011010)_2+(001100)_2$ **ii.** $(0011010)_2-(001100)_2$	L-5	CO6	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Describe the process of minimizing a logic expression in digital logic circuits using K-Map.	L-2	CO1	**10**
		OR			
	b.	Describe the different types of flip-flops, including D, T, JK, and SR flip-flops.	L-2	CO1	**10**
9.	**a.**	Add the binary numbers 100111 and 11011 Multiply the binary numbers 1010 and 1001	L-5	CO6	**5** **5**
		OR			
	b.	(i)Explain Instruction Cycle with example. (ii)Explain Binary Addition, Subtraction with example.	L-3 L-3	CO2 CO2	**5** **5**
10.	**a.**	Illustrate the booth multiplication flow chart in detail.	L-5	CO3	**10**
		OR			
	b.	Explain CISC instruction set with its characteristics, Advantages and Disadvantages.	L-4	CO3	**10**
11.	**a.**	Explain Daisy-Chaining Priority interrupt Method.	L-3	CO4	**10**
		OR			
	b.	Explain Flynn's classification with detail discussion.	L-3	CO4	**10**
12.	**a.**	Describe the different types of Replacement Algorithms in detail.	L-4	CO5	**10**
		OR			
	b.	Discuss in detail about Memory Management.	L-3	CO5	**10**
13.	**a.**	Simplify using a K-map with sum of products for the Boolean function $Y(A, B, C, D) = \Sigma m (0,1,2,3,5,7,8,9,11,14)$ and draw the minimized logic diagram.	L-5	CO1	**10**
		OR			
	b.	Convert the following **i.** Convert $(10011)_2$ to a decimal number. **ii.** Binary Subtract $(11010)_2$ and $(10110)_2$ **iii.** Binary multiplication $1010_2 \times 101_2$	L-5	CO6	**10**